Dynamics 4

Past paper solutions

Note These are examples of numerical solutions for the Dynamics 4 exam papers published online since 2013-14. You can find the exam papers on this link. Please remember that most solutions are not unique and depend on assumptions such as coordinate systems, references, *etc.* Non-numerical answers are not included.

Exam date: 13-December-2013

Question 1

b) $Q_y = -c[\dot{y} + \omega Z \sin(\omega t)], \quad Q_\varphi = -\tau$
c) $M\ddot{y} + mr(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2 \sin\varphi) + k[y - Z\cos(\omega t)] = -c[\dot{y} + \omega Z \sin(\omega t)]$
$J\ddot{\varphi} + mr\ddot{y}\cos\varphi = -\tau$

Question 2

a) $\varphi_\mathrm{a}(x) = \left(\frac{x}{2L}\right)^2, \quad \varphi_\mathrm{b}(x) = 1 - \cos\left(\frac{\pi x}{4L}\right)$
b) $\omega_\mathrm{a}^2 = 0.625\frac{EI}{mL^3}, \quad \omega_\mathrm{b}^2 = 0.84\frac{EI}{mL^3}$
d) $Q_\mathrm{a} = \frac{17}{48} f_0 L = 0.354 f_0 L$

Question 3

a) $\omega_1 = 96.1$ rad/s, $\quad \omega_2 = 309.2$ rad/s
b) $\lambda_1 = 4L = 4.0$ m, $\quad \lambda_2 = 2L = 2.0$ m

Exam date: 12-December-2014

Question 1

a) $y(t)$ (mass), $\quad \theta(t)$ (bar)
c) $Q_y = 0, \quad Q_\theta = -c\left(\frac{L^2}{9}\dot{\theta}\cos\theta\right)$
d) $M\ddot{y} + \frac{1}{2}ky - \frac{1}{2}kL\theta + Mg = 0,$ with $k = k_2 = 2k_1$
$\frac{1}{3}mL^2\ddot{\theta} + \frac{11}{18}kL^2\theta - \frac{1}{2}kLy + \frac{1}{2}mgL = -\frac{1}{9}cL^2\dot{\theta},$ with $k = k_2 = 2k_1$

Question 2

a) $\varphi_1(x) = 1, \quad \varphi_2(x) = \cos\left(\frac{\pi x}{L}\right)$
c) $Q_1 = \frac{1}{4} f_0 L\cos(\omega t), \quad Q_2 = \frac{4L}{\pi^2}(2-\sqrt{2}) f_0 \cos(\omega t)$
d) $m\ddot{q}_1 + \frac{2}{\pi}m\ddot{q}_2 + 2kq_1 + mg = \frac{1}{4} f_0 L\cos(\omega t)$
$\frac{2m}{\pi}\ddot{q}_1 + \frac{m}{2}\ddot{q}_2 + \frac{\pi^4 EI}{2L^3}q_2 + \frac{2}{\pi}mg = \frac{4L}{\pi^2}(2-\sqrt{2}) f_0 \cos(\omega t)$

Question 3

a) $\frac{1}{3}ml^2\ddot{\varphi} - \left(\frac{1}{2}mgl - \frac{10}{9}kl^2\right)\varphi = 0$
b) Unstable if $k < \frac{27}{60}\frac{mg}{l}$

Exam date: 17-December-2015

Question 1

a) $y(t)$ (mass), $\quad \theta(t)$ (bar)
c) $Q_y = 0, \quad Q_\theta = -c\left(\frac{L^2}{9}\dot{\theta}\cos\theta\right)$
d) $M\ddot{y} + ky - \frac{2}{3}kL\theta + Mg = 0$ and $\frac{1}{9}mL^2\ddot{\theta} + \frac{5}{9}kL^2\theta - \frac{2}{3}kLy = -\frac{1}{9}cL^2\dot{\theta}$

Question 2

a) $\frac{1}{3}mh^2\ddot{\theta} - \frac{1}{2}mgh\theta + k\Delta^2\theta = -c\Delta^2\dot{\theta}$
b) Unstable if $k < 1226.3$ MN/m

Question 4

a) $\varphi_1(x) = 1, \quad \varphi_2(x) = \sin\left(\frac{\pi x}{L}\right)$
b) $Q_1 = \frac{L}{4} f_0 \cos(\omega t), \quad Q_2 = \frac{\sqrt{2}}{2}\frac{f_0 L}{\pi}\cos(\omega t)$
c) $m\ddot{q}_1 + \frac{2m}{\pi}\ddot{q}_2 + 2kq_1 + mg = \frac{1}{4} f_0 L \cos(\omega t)$ and $\frac{2m}{\pi}\ddot{q}_1 + \frac{m}{2}\ddot{q}_2 + \frac{2mg}{\pi} + \frac{\pi^4 EI}{2L^3} q_2 = \frac{\sqrt{2}}{2}\frac{f_0 L}{\pi}\cos(\omega t)$

Exam date: 19-December-2016

Question 1

a) $y(t)$ (mass), $\theta(t)$ (bar)
c) $Q_y = 0, \quad Q_\theta = -ca^2\dot{\theta}\cos\theta$
d) $M\ddot{y} + k_2(y - L\theta) - Mg = 0$ and $\frac{1}{3}mL^2\ddot{\theta} - \frac{1}{2}mgL + k_1 b^2\theta - k_2 L y + k_2 L^2\theta = -ca^2\dot{\theta}$

Question 2

a) $I_A\ddot{\theta} - \frac{1}{3}Mgr\sin\theta = -c\frac{d^2}{4}\dot{\theta}\cos\theta$
b) The system is unstable, regardless of the damping

Question 4

a) $\varphi(x) = \left(\frac{x}{L}\right)^2, \quad \varphi(x) = 1 - \cos\left(\frac{\pi x}{2L}\right)$
b) $\frac{1}{5}\rho AL\ddot{q} + \frac{4EI}{L^3} q = 0.09 f_0 L \cos\omega t$
c) Pine wood not good because elastic modulus is $E > 3$ GPa

Exam date: 21-December-2017

Question 1

a) $y(t)$ (mass), $\theta(t)$ (rotation of blades)
c) $Q_y = -c\dot{y}, \quad Q_\theta = 0$
d) $(M - m - m')\ddot{y} + Mg + 2ky = -c\dot{y}$ and $\left[mr^2 + m'(r')^2\right]\ddot{\theta} + g(mr - m'r')\cos\theta = 0$

Question 2

a) $\omega_1 = \frac{0.879}{L^2}\sqrt{\frac{EI}{\rho A}}$
b) $\varphi_1(x) = \frac{x^2}{4L^2}, \quad 1 - \cos\left(\frac{\pi x}{4L}\right)$
c) $\omega_{n1}^2 = 1.25\frac{EI}{mL^3}, \quad \omega_{n2}^2 = 0.84\frac{EI}{mL^3}$
d) $Q_1 = f_0 L\left(\frac{15}{4} - \frac{7}{3}\right) = 1.417 f_0 L$

Question 3

a) $\frac{1}{2}ml^2\ddot{\theta} - \frac{1}{2}ml^2\omega^2\cos\theta\sin\theta + mgl\sin\theta = 0$
b) $\theta^{\mathrm{s}} = \pi$ always unstable
$\theta^{\mathrm{s}} = 0$ unstable if $\omega^2 > 2g/l$

Exam date: 12-December-2018

Question 1

a) $y(t)$ (vertical displacement), $\theta(t)$ (rotation of blades)
c) $Q_y = -c\dot{y}, \quad Q_\theta = 0$
d) $(m_1 + 2m_2)R^2\ddot{\varphi} + (m_2 - m_1)gR\sin\varphi = 0$ and $M\ddot{y} + 2ky + (M + m_1 + 2m_2)g = -c\dot{y}$

Question 2

a) $(M + m/3)l^2\ddot{\theta} - (M + m/2)gl\sin\theta = 0$
b) Unstable
d) $(M + m/4)l^2\ddot{\theta} + (M + m/2)l(\ddot{\bar{z}}\cos\theta - g\sin\theta) = 0$

Question 3

a) $\varphi_{\mathrm{a}}(x) = \sin(\pi x/L)$ or $\varphi_{\mathrm{b}} = 16x(x^3 - 2Lx^2 + L^3)/(5L^3)$
b) $h = 6$ mm
c) 16%
d) $A = 0.37$ mm

Exam date: 11-December-2019

Question 1

a) $\frac{1}{3}\left(\rho_1 L_1^3 + \rho_2 L_2^3\right) A\ddot{\varphi} - \frac{1}{2}\left(\rho_2 L_2^2 - \rho_1 L_1^2\right) Ag\sin\varphi = 0$
b) $L_1 \geq 0.59 L_2$
c) $\omega = 3.04$ rad/s (0.48 Hz)

Question 2

→ Note: dry-weight of wing is $m = 4400$ kg
a) $\varphi_{\mathrm{a}}(x) = \left(\frac{x}{L}\right)^3$ and $\varphi_{\mathrm{b}}(x) = \left[1 - \cos\left(\frac{\pi x}{2L}\right)\right]$
b) $T = \left[\frac{1}{2}M\left(\frac{a}{L}\right)^6 + \frac{1}{14}m\right]\dot{q}^2$ and $V = \left[M\left(\frac{a}{L}\right)^3 + \frac{m}{4}\right]gq + \frac{6EI}{L^3}q^2$
c) $\left[M\left(\frac{a}{L}\right)^6 + \frac{m}{7}\right]\ddot{q} + \frac{12EI}{L^3}q + \left[M\left(\frac{a}{L}\right)^3 + \frac{m}{4}\right]g = 0$
d) $a = 8.2$ m

Question 3

a) $\varphi(x) = \left[1 - \cos\left(\frac{\pi x}{2L}\right)\right]$
b) $\omega = 152.8$ rad/s (24.3 Hz)
c) $\bar{\omega} = 123.8$ rad/s (19.7 Hz, 19% decrease in the wave frequency)
d) $A_1 = 12.9$ cm

Exam date: 18-December-2020

Question 1

a) $\frac{14}{3}m_1 h_1^2\ddot{\theta} + 2kh_1^2\sin\theta\cos\theta\left(\frac{1}{\sin^2\alpha_1} + \frac{4}{\sin^2\alpha_2}\right) - \frac{5}{2}m_1 g h_1\sin\theta = 2F_1(h_1 + h_2)\cos\theta$
c) $k = 80.4$ kN/m

Question 2

a) $\varphi_{\mathrm{a}}(x) = \left(\frac{2x}{L}\right)^2 - 1$ and $\varphi_{\mathrm{b}}(x) = \cos\left(\frac{\pi x}{L}\right)$
b) $\left(M\Omega_a + \frac{8}{15}m\right)\ddot{q} + \frac{64EI}{L^3}q = -\frac{13}{81}Lg_0\cos(\omega t)$ with $\Omega_a = \left(\frac{2t}{L}\right)^4 - 2\left(\frac{2t}{L}\right)^2 + 1$
c) $M = 0.187$ kg (187 g)

Question 3

a) $\omega = 103.9$ rad/s (16.5 Hz)
b) $\bar{\omega} = 82.2$ rad/s (13.1 Hz, 21% decrease in the wave frequency)
c) $A_1 = 45.7$ cm

Exam date: 17-December-2021

Question 1

a) $M\ddot{x} + k_1\left(x - \frac{2L}{3}\theta\right) + k_2 x = 0$ and $mL^2\ddot{\theta} + 6k_1 L(\theta - x) = -cL^2\dot{\theta}$
b) $k_1 = 3.95 \times 10^{-2}$ N/m and $k_2 = 15.6$ N/m
c) $T = \frac{1}{2}M\dot{x}^2 + \frac{1}{18}mL^2\dot{\theta}^2 + \frac{1}{2}m\left(\frac{L}{3}\dot{\theta}\dot{z}\cos\theta + \dot{z}^2\right)$ and $V = \frac{1}{2}k_1\left(x - \frac{2L}{3}\theta - \bar{z}\right)^2 + \frac{1}{2}k_2 x^2$

Question 2

a) $\varphi_1(x) = \sin\left(\frac{\pi x}{L}\right)$ and $\varphi_2(x) = \sin\left(\frac{2\pi x}{L}\right)$
b) $\left(\frac{3M}{4} + \frac{m}{2}\right)\ddot{q} + \frac{\pi^4 EI}{2L^3}q = \frac{f_0 L}{\pi^2}\cos(\omega t)$
c) $M = 139.3$ kg

Question 3

a) $\omega = 194.7$ rad/s (31 Hz)
b) The mas is 0.85 kg and the impact velocity is 6.1 m/s
c) $\omega = 154.5$ rad/s $= 24.6$ Hz (15.2% reduction)

Exam date: 10-December-2022

Question 1

a) $x(t)$ (vertical displacement), $\theta(t)$ (rotation of bar)
b) $T = \frac{1}{2}M\dot{x}^2 + \frac{1}{18}mL^2\dot{\theta}^2$ and $V = Mgx - \frac{1}{6}mgL\cos\left(\frac{\pi}{6} - \theta\right) + \frac{1}{2}k\left(x - \frac{1}{3}L\sin\theta\right)^2$
c) $Q_x = 0$ and $Q_\theta = -c\left(\frac{L}{3}\right)^2\dot{\theta}\cos\theta$
d) $M\ddot{x} + kx - \frac{1}{3}kL\theta = 0$ and $\frac{1}{9}mL^2\ddot{\theta} + \frac{1}{3}kL\left(\frac{1}{3}L\theta - x\right) = -c\frac{L^2}{9}\dot{\theta}$

Question 2

a) For $\theta^s = 0$ stable if $\frac{L_1}{L_2} \geq \frac{M_2 + \frac{m_2}{2}}{M_1 + \frac{m_1}{2}}$, for $\theta^s = \pi$ stable if $\frac{L_1}{L_2} \leq \frac{M_2 + \frac{m_2}{2}}{M_1 + \frac{m_1}{2}}$
b) For $\theta^s = 0$ stable if $\frac{L_1}{L_2} \geq 1.5$, for $\theta^s = \pi$ stable if $\frac{L_1}{L_2} \leq 1.5$

Question 3

a) $\omega = 2.5$ rad/s
b) $\bar{\omega} = 2.25$ rad/s and $A = 0.3$ m
c) $c = 2.8$ m/s

Exam date: 15-December-2023

Question 1

a) $y(t)$ (vertical linear motion), $\theta(t)$ (angular motion)
b) $T = \frac{1}{2}(M+m)\dot{y}^2 + \frac{1}{6}mL^2\dot{\theta}^2 - \frac{1}{2}mL\dot{y}\dot{\theta}\sin\theta$ and $V = \frac{1}{2}ky^2 - \left(My + \frac{1}{2}mL\cos\theta + my\right)g$
c) $Q_y = -c\dot{y}$ and $Q_\theta = -LP_0\cos(\omega t)\cos\theta$
d) $(M+m)\ddot{y} + ky - \frac{1}{2}mL\ddot{\theta}\theta - (M+m)g = -c\dot{y}$ and $\frac{1}{3}mL^2\ddot{\theta} - \frac{1}{2}mL\ddot{y}\theta + \frac{1}{2}mgL\theta = -LP_0\cos(\omega t)$

Question 2

a) $\frac{1}{2}\left(\frac{M}{2} + m\right)\ddot{q} + \frac{\pi^4 EI}{2L^3}q = 0$
b) $M = 51$ g
c) $\Delta\omega = 3.3\%$, with $\bar{\omega} = 2672$ rad/s ($\bar{\omega} = 425$ Hz)
d) $v_0 = 3.78$ m/s

Question 3

a) $\varphi(x) = 1 - \cos\left(\frac{\pi x}{2h}\right)$
b) $Q_q = 1.394 f_0\cos(\omega t)$
c) $0.226m\ddot{q} + 3.044\frac{EI}{h^3}q = 1.394 f_0\cos(\omega t)$
d) $t = 0.2$ m

Exam date: 12-December-2024

Question 1

a) $T = \frac{3}{2}ML^2\dot{\theta}^2$ and $V = \frac{1}{8}kL^2\sin^2\theta + 4MgL\cos\theta$
b) $3ML^2\ddot{\theta} + \frac{1}{4}kL^2\sin\theta\cos\theta - 4MgL\sin\theta = -c\frac{L^2}{4}\dot{\theta}\cos\theta$
c) Stable if the stiffness of the spring is greater than $16Mg/L$

Question 2

a) $\varphi(x) = \left(\frac{x}{L}\right)^2$
b) $T = \frac{1}{10}m\dot{q}^2$ and $V = \frac{1}{3}mgq + \frac{2EI}{L^3}q^2$
c) $\frac{1}{5}m\ddot{q} + \frac{4EI}{L^3}q + \frac{1}{3}mg = 0$
d) $EI = 1.73 \times 10^9$ Nm^2
e) Frequency reduction $\approx 0.9\%$ and $A = 1$ mm

Question 3

a) $\varphi(0) = 0$, $\varphi'(0) = 0$, $\varphi(L) = 0$ and $\varphi''(L) = 0$
b) $\omega_1 = 15.4\sqrt{EI/(\rho AL^4)}$ and $\omega_2 = 50.0\sqrt{EI/(\rho AL^4)}$
c) A soft wood with an elastic modulus of 8 GPa could not be used as it does not comply with the design requirement

Exam date: 12-December-2025

Question 1

a) Vertical displacement of mass M, $y(t)$, and (ii) the angular rotation of the rod AB, $\theta(t)$

b) $T = \frac{1}{2}(M+m)\dot{y}^2 + \frac{1}{8}mL^2\dot{\theta}^2 + \frac{1}{2}mL\dot{\theta}\dot{y}\cos\theta$
$V = ky^2 + \frac{1}{2}kL^2\sin^2\theta + kLy\sin\theta + (M+m)gy + \frac{1}{2}mgL\sin\theta$

c) $Q_\theta = 0 \quad \text{and} \quad Q_y = -c\dot{y}$

d) $(M+m)\ddot{y} + \frac{1}{2}mL\ddot{\theta} + (M+m)g + 2ky + kL\theta = -c\dot{y}$
$\frac{1}{4}mL^2\ddot{\theta} + \frac{1}{2}mL\ddot{y} + kL^2\theta + kLy + \frac{1}{2}mgL = 0$

Question 2

a) $u'(0,t) = 0$ and $u'(L,t) = -\frac{M}{EA}\ddot{u}(L,t)$

b) $\omega_1 = 4395$ rad/s $= 699$ Hz and $\omega_2 = 9300$ rad/s $= 1480$ Hz

c) $\lambda_1 = 5.03$ m and $\lambda_2 = 2.38$ m

Question 3

a) $\varphi_1(x) = \sin\left(\frac{\pi x}{L}\right)$ and $\varphi_2(x) = \sin\left(\frac{2\pi x}{L}\right)$

b) $\frac{1}{2}\rho AL\ddot{q}_1 + \frac{\pi^4 EI}{2L^3}q_1 = 0$

c) $\omega_1 = \sqrt{\frac{\pi^4 EI}{mL^3}}$

d) Minimum diameter is $d = 0.27$ m $= 27$ cm.

e) $\bar{\omega}_1 = 12.5$ rad/s $= 1.99$ Hz and $\Delta\omega_1 = 0.07$ rad/s $= 0.01$ Hz. Post-impact amplitude of vibration is $\bar{A}_1 = 7$ mm.